

ZHAO HONGMING

Singapore PR | hi@xploreurself.com | xploreurself.com | github.com/hongming-github

SUMMARY

Software engineer with 9 years building production backend systems in regulated banking, now pursuing a master's degree in AI systems and building multi-agent LLM systems end-to-end. I work hands-on with LangGraph and LangChain, wrapping existing services as tools behind a ReAct agent, routing intent across independently-deployable pipelines, and enforcing structured output at every boundary. I care about making agent behavior measurable, so I build evaluation harnesses with labelled ground truth, multi-model comparison, and statistical testing instead of relying on subjective judgement.

PERSONAL PROJECT

Job Hunt Agent

- Built and operate a multi-agent job-hunting pipeline that scrapes 2,000+ listings daily, scores them against my resume with an LLM, and pushes ranked matches to a Telegram bot with async CV generation.
- Exposed the system as a FastMCP server (6 tools + 2 resources) consumable from Claude Desktop.
- Built a feedback loop that turns natural-language user feedback into structured rules, applying soft scoring penalties for vague preferences and hard SQL filters for explicit exclusions.
- Diagnosed unstable OpenAI prompt cache hits down to load-balancer shard routing and raised the hit rate from 47.7% to 95.4% with per-worker cache keys.
- Cut output token cost 61% by trimming unused fields from the scoring schema, verified through per-batch cost logging.

Tech: Python, LangGraph, LangChain, OpenAI API, FastMCP, Redis, PostgreSQL, Docker, python-telegram-bot, cron

WORK EXPERIENCE

AI Engineer Intern — ST Engineering — Mar 2026 – Present

Multi-Agent System for Enterprise Use Cases

- Built a multi-agent system that lets non-technical users drive 6 existing AI microservices through natural language, wrapping standalone models as LangChain tools behind a ReAct agent for multi-step reasoning and tool composition.
- Designed a LangGraph architecture with an LLM intent router across 6 independently-deployable pipelines, validated first as a Langflow PoC to de-risk feasibility before committing to a build.
- Built a versioned evaluation harness with labelled ground-truth sets, faithfulness scoring, and multi-model comparison.
- Ran a two-tier router experiment escalating low-confidence cases to a stronger model; it delivered no accuracy gain at 3.5x cost with worse calibration, so I traced the cause to the strong model's lower abstention bias and rolled the change back.
- Designed prefix-cache-compatible prompt templates to reduce token cost on supporting providers.

Tech: Python, Langflow, LangGraph, LangChain, LangFuse, FastAPI, Amazon Bedrock, Redis, PostgreSQL, Docker, Streamlit

Tech Lead — DBS Bank — Oct 2024 – Aug 2025

Wealth Domain · Pre-Trade Check Engine

- Led a 4-person onshore/offshore team delivering a pre-trade check engine for wealth products across 4 regional banking systems, owning delivery end-to-end from requirements to production.
- Translated ambiguous business and regulatory requirements into technical specs, working directly across security, business, and downstream domain teams to align on functional and non-functional needs.
- Owned the architecture: Drools rule modelling, Spring Boot microservices, Kafka event flow, MariaDB, PCF deployment.
- Provided cost / timeline estimation and coordinated production release planning across multiple stakeholder groups.

Tech: Java, Spring Boot, Drools, Microservices, MariaDB, Pivotal Cloud Foundry, Kafka, Grafana

Senior Backend Engineer — Trust Bank Singapore — Apr 2023 – Sep 2024

Customer & Onboarding Domain

- Delivered customer and onboarding projects using event-driven architecture to decouple components and improve system scalability.
- Built monitoring and data dashboards on internal logging tooling to surface business insights on newly released features and aid production support.
- Mentored junior developers, ran code reviews to maintain code quality, and partnered with L1 / L2 support on production troubleshooting.

Tech: Kotlin / Java, Spring Boot, Microservices, PostgreSQL, AWS, Kafka, Terraform, SumoLogic

API Developer — OCBC Bank — Mar 2019 – Mar 2023

OCBC Enterprise API

- Acted as primary technical contact onboarding dozens of enterprise partners including tech, government, education, and fintech partners across 4 regions onto the OCBC Enterprise API platform, owning the relationship from integration through go-live.
- Collaborated with business, security, and IT teams to develop cross-regional public API services while ensuring compliance with security standards.
- Coordinated with infrastructure, network, and operations teams to stand up new environments across regions; provided L3 support on live-environment issues.

Tech: Java, Spring Boot, API Gateway, Microservices, PostgreSQL, Kafka, OpenShift, ELK

Software Engineer — Avanade — Jul 2017 – Feb 2019

Government Project

- Developed RESTful web services using Spring and Spring Data JPA.
- Integrated Spring-based services with upstream and downstream systems across teams, covering data contracts, error handling, and release coordination.

Tech: Java, Spring, Maven, CI/CD, Splunk

API Developer Intern — OCBC Bank — Oct 2016 – Mar 2017

OCBC Open API Platform (api.ocbc.com)

- Enhanced third-party developer access to OCBC services and data, improving collaboration with new enterprise partners.
- Refactored front-end code from static to dynamic pages and customised the OAuth2 flow to fit partner requirements.

Tech: Java, Spring, Maven, jQuery, Jaggery.js, Bootstrap

TECHNICAL SKILLS

AI / LLM: LangGraph, LangChain, ReAct agents, multi-agent orchestration, tool-calling / function calling, structured output validation, LLM evaluation, RAG, Amazon Bedrock, MCP, prompt caching, Claude Code

Backend: Spring Boot, FastAPI, microservices, Kafka, event-driven architecture, REST / API Gateway, Drools

Languages: Java / Kotlin, Python, JavaScript, C#

Data / Infra: PostgreSQL, MariaDB, Docker, Kubernetes, AWS, PCF, OpenShift, Terraform, CI/CD, Git

EDUCATION

National University of Singapore — Master of Technology, Artificial Intelligence Systems (2025 – Present)

National University of Singapore — Graduate Diploma in Systems Analysis (2016 – 2017)

Zhejiang University of Technology — B.Eng. Computer Science (Digital Media Technology) (2012 – 2016)